

Project 1: Realtime AI Medical Research & Synthesis

Crew – Agentic AI with crew.ai

This project demonstrates an agentic AI system built with CrewAI that assists a medical research team. It combines real-time information gathering (news, research papers), retrieval from a proprietary knowledge base (simulated), and the reasoning capabilities of Amazon Bedrock LLMs to provide comprehensive, up-to-date, and contextually rich insights on diseases and treatments.

Core Concepts Demonstrated:

- **Agentic AI with CrewAI:** Orchestrating specialized AI agents to collaborate on complex tasks.
- **Inter-Agent Communication:** Agents passing refined information and delegating sub-tasks.
- **Context Management:** How CrewAI manages context across agents and tasks.
- **Tool Usage:** Leveraging pre-built (SerperDevTool, ScrapeWebsiteTool) and custom tools (Proprietary KB Retriever).
- **Amazon Bedrock Integration:** Utilizing powerful foundation models (e.g., Anthropic Claude, Amazon Titan) for agent intelligence.
- **Real-time Information Integration:** Pulling live data from the web.
- **Proprietary Knowledge Integration:** Accessing internal, domain-specific information.

Use Case: "Accelerating Medical Research for Rare Diseases"

Scenario: A specialized medical research lab, "Synapse Innovations," is focused on finding new treatments for rare neurological disorders. Researchers often spend days or weeks sifting through new scientific publications, clinical trial updates, and internal research data to identify potential breakthroughs or refine existing hypotheses. They need a system that can rapidly synthesize information from diverse sources, prioritize relevant findings, and present actionable insights.

Challenges this demo addresses (and Agentic AI improves):

1. **Information Overload:** The sheer volume of new medical literature makes it impossible for humans to keep up.
2. **Disconnected Data Silos:** Information is fragmented across public databases, real-time news, and internal research repositories.

3. **Synthesis Complexity:** Extracting critical connections and drawing conclusions from disparate data points is time-consuming and prone to human bias/oversight.
4. **Real-time Relevance:** The need to incorporate the very latest findings immediately.
5. **Contextual Depth:** Combining broad scientific understanding with specific proprietary research data.

Project Structure & Implementation Details:

1. Setup

- **AWS Credentials:** Ensure your AWS credentials (Access Key ID, Secret Access Key, Region Name) are configured (e.g., via environment variables or `~/.aws/credentials`).
- **Amazon Bedrock Model Access:** Request access to the desired models in your AWS Bedrock console (e.g., Anthropic Claude 3 Sonnet, Amazon Titan Text Express).
- **Serper API Key:** Obtain an API key from [Serper.dev](https://serper.dev) for real-time search.
- **Install Libraries:**

```
pip install -q crewai crewai_tools boto3 langchain_aws
```

2. AWS Bedrock LLM Configuration

```
import os
```

```
from crewai import Agent, Task, Crew, Process
```

```
from crewai_tools import SerperDevTool, ScrapeWebsiteTool
```

```
from langchain_aws import ChatBedrock # For Chat models like Claude
```

```
from langchain_aws import BedrockLLM # For Text Completion models like Titan
```

```
# Set environment variables for AWS Bedrock and Serper
```

```
os.environ["AWS_ACCESS_KEY_ID"] = "YOUR_AWS_ACCESS_KEY_ID"
```

```
os.environ["AWS_SECRET_ACCESS_KEY"] = "YOUR_AWS_SECRET_ACCESS_KEY"
```

```
os.environ["AWS_REGION_NAME"] = "YOUR_AWS_REGION" # e.g., "us-east-1"
```

```
os.environ["SERPER_API_KEY"] = "YOUR_SERPER_API_KEY"
```

```
# Choose your Bedrock LLM. Claude 3 Sonnet is excellent for agents.

# You can also use Amazon Titan Text Express or others.

# Ensure you have access to the chosen model in Bedrock.

bedrock_llm_model_id = "anthropic.claude-3-sonnet-20240229-v1:0" # Or
"amazon.titan-text-express-v1"

bedrock_llm = ChatBedrock(

    model_id=bedrock_llm_model_id,

    region_name=os.environ["AWS_REGION_NAME"],

    temperature=0.2, # Lower temperature for factual/research tasks

    max_tokens=2048 # Adjust as needed

)
```

3. Proprietary Knowledge Base Tool (Simulated)

In a real scenario, this would connect to a database (e.g., AWS Aurora with PGVector, Amazon Kendra, or a custom vector store). Here, we simulate it.

```
from crewai_tools import BaseTool

import json
```

```
# Simulate a proprietary knowledge base (e.g., internal research papers, clinical trial
results)
```

```
# In a real system, this would be a database query or RAG system.
```

```
PROPRIETARY_KB_DATA = {

    "ALS_experimental_drug_X": {

        "title": "Phase 2 Clinical Trial Results: Experimental Drug X for ALS",

        "authors": "Dr. A. Sharma, Dr. B. Lee",

        "date": "2024-03-15",
```

"summary": "Drug X showed promising results in slowing disease progression in 30% of ALS patients in a Phase 2 trial. Mild gastrointestinal side effects observed. Further research into biomarker response needed.",

"findings": "Significant reduction in ALSFRS-R decline rate in subgroup. No severe adverse events. Mechanism of action involves enhancing mitochondrial function.",

"related_studies": ["Pubmed:12345", "Internal_Project:P001"]

},

"Alzheimers_biomarker_Y": {

"title": "Novel CSF Biomarker Y for Early Alzheimer's Detection",

"authors": "Dr. C. Patel",

"date": "2023-11-01",

"summary": "Biomarker Y in cerebrospinal fluid demonstrates high specificity and sensitivity for early-stage Alzheimer's disease, even before cognitive decline. Potential for non-invasive blood test under investigation.",

"findings": "Correlation with amyloid plaque burden confirmed. Longitudinal study ongoing for predictive value.",

"related_studies": ["Nature:67890", "Internal_Review:R123"]

},

"Parkinsons_gene_therapy_Z": {

"title": "Pre-clinical Efficacy of Gene Therapy Z for Parkinson's Disease",

"authors": "Dr. D. Wong",

"date": "2025-01-20",

"summary": "Gene therapy Z, targeting alpha-synuclein aggregation, showed significant neuroprotective effects in mouse models of Parkinson's. Human trials planned for late 2025.",

"findings": "Reduced alpha-synuclein pathology and improved motor function in animal models. Delivery method optimization is key.",

"related_studies": []

}

}

```

class ProprietaryKBRetriever(BaseTool):

    name: str = "Proprietary Knowledge Base Retriever"

    description: str = "A tool to retrieve detailed information from Synapse Innovations' internal research database. Input should be a specific research topic, disease name, or experimental drug name (e.g., 'ALS_experimental_drug_X', 'Alzheimers_biomarker_Y')."

    def _run(self, query: str) -> str:
        """
        Simulates retrieving information from a proprietary knowledge base.

        In a real-world scenario, this would involve a database query, RAG, etc.
        """

        # Simple lookup in our simulated KB

        data = PROPRIETARY_KB_DATA.get(query.replace(" ", "_"), None)

        if data:

            return json.dumps(data, indent=2)

        else:

            return f"No specific information found in the proprietary knowledge base for '{query}'. Please try a more specific query or broad search."

proprietary_kb_tool = ProprietaryKBRetriever()

# Initialize CrewAI's pre-built tools
serper_tool = SerperDevTool()
scrape_website_tool = ScrapeWebsiteTool()

```

4. Define Agents

Each agent has a clear role, goal, backstory, and tools.

Below is a sample implementation (augment to create your own)

We will use the supervisor > Worker pattern (Explore other agentic ai patterns and put them in your presentation)

The Supervisor will review the Draftsman's output. If it identifies areas needing deeper clarification, better phrasing, or specific recommendations, it can delegate *back* to the Draftsman with specific instructions for refinement. This demonstrates iterative improvement and quality control.

4 Agents:

Agent 1: Clinical Literature Specialist

Agent 2: Internal Research Analyst

Agent 3: Report Draftsman (Worker)

Agent 4: Synthesizer (Supervisor)

Sample implementation below:

Agent 1: Clinical Literature Specialist

```
clinical_specialist = Agent(  
    role='Clinical Literature Specialist',  
    goal='Identify the latest significant research, clinical trials, and breakthrough news  
related to specific medical conditions or treatments.',  
    backstory=(  
        "You are a highly experienced Clinical Literature Specialist at Synapse Innovations. "  
        "Your expertise lies in rapidly scanning vast amounts of new medical publications, "  
        "clinical trial registries, and healthcare news sources. You are meticulous "  
        "in identifying peer-reviewed studies, impactful reviews, and emerging trends "  
        "that could inform our research. You prioritize credibility and relevance."  
    ),  
    llm=bedrock_llm,  
    tools=[serper_tool, scrape_website_tool],  
    verbose=True,  
    allow_delegation=False  
)
```

Agent 2: Internal Research Analyst

```
internal_analyst = Agent(  
    role='Internal Research Analyst',  
  
    goal='Cross-reference external findings with Synapse Innovations\' proprietary  
research data and identify internal connections or discrepancies.',  
  
    backstory=(  
        "You are a dedicated Internal Research Analyst for Synapse Innovations, "  
        "with deep knowledge of our internal clinical trials, preclinical studies, "  
        "and proprietary datasets. Your task is to take external insights from the "  
        "Clinical Literature Specialist and check them against our confidential "  
        "knowledge base, highlighting any direct matches, conflicting results, "  
        "or synergistic opportunities."  
    ),  
  
    llm=bedrock_llm,  
  
    tools=[proprietary_kb_tool],  
  
    verbose=True,  
  
    allow_delegation=True # This agent can delegate if it needs more specific info  
)
```

Agent 3: Report Draftsman (Worker)

```
report_draftsman = Agent(  
    role='Medical Report Draftsman',  
  
    goal='Prepare clear, structured first drafts of medical research summaries based on  
provided external and internal findings.',  
  
    backstory=(  
        "You are a skilled Medical Report Draftsman specializing in scientific  
communication. "  
  
        "Your task is to take raw research findings and internal analysis, and transform "
```

```
"them into a well-organized, coherent, and preliminary draft of a research summary.
"
```

```
"You focus on clarity, accuracy, and completeness for the initial outline."
```

```
),
llm=bedrock_llm,
verbose=True,
allow_delegation=False # This agent focuses on drafting
)
```

```
# Agent 4: Research Insights Synthesizer (Now the Supervisor)
```

```
insights_synthesizer_supervisor = Agent(
    role='Senior Research Insights Supervisor',
    goal='Oversee and refine the synthesis of all gathered information, ensuring the final
output is of the highest quality, actionable, and aligns with research team needs.',
    backstory=(
        "You are the Senior Research Insights Supervisor at Synapse Innovations. "
        "Your critical eye ensures that all research summaries are precise, impactful, "
        "and directly useful for advancing medical research. You review drafts, identify "
        "areas for improvement or deeper analysis, and guide junior agents (like the "
        "Medical Report Draftsman) to produce a perfect final report. You are the "
        "ultimate arbiter of quality and completeness."
    ),
    llm=bedrock_llm,
    verbose=True,
    allow_delegation=True # Crucial for delegation to worker agents
)
```

5. Define Tasks

Tasks describe the specific actions agents need to perform and how they communicate.

4 Tasks:

Task 1: Research Latest Clinical News

Task 2: Cross-reference with Proprietary KB

Task 3: Draft Initial Research Summary (Worker Task)

Task 4: Supervisor Review and Finalization

Sample implementation below:

Task 1: Research Latest Clinical News

```
task_research_news = Task(  
    description=(  
        "Search for the latest research papers, clinical trial updates, and news articles "  
        "on the topic: '{medical_query}'. "  
        "Focus on publications from the last 6 months. "  
        "Identify 3-5 key findings or developments."  
        "The output should be a detailed list of findings with source URLs."  
    ),  
    expected_output="A bulleted list of 3-5 key recent research findings related to  
'{medical_query}', each with a brief summary and a source URL.",  
    agent=clinical_specialist  
)
```

Task 2: Cross-reference with Proprietary KB

```
task_cross_reference_kb = Task(  
    description=(  
        "Given the external research findings on '{medical_query}', "  
        "query Synapse Innovations' internal proprietary knowledge base. "  
        "Identify if we have existing research, clinical trial data, or "  
        "pre-clinical results related to these external findings. "  
        "Specifically look for direct matches, contradicting evidence, "  
        "or complementary insights. If the external findings mention "
```

```

        "specific drugs or biomarkers, try to find those in our KB."
        "The output should highlight internal connections and discrepancies, "
        "along with any relevant details from our proprietary KB."
    ),
    expected_output=(
        "A summary of internal knowledge base findings relevant to '{medical_query}', "
        "highlighting any direct matches, contrasting data, or synergistic points with "
        "the external research. State clearly if no direct internal match is found."
    ),
    agent=internal_analyst,
    context=[task_research_news]
)

```

Task 3: Draft Initial Research Summary (Worker Task)

```

task_draft_summary = Task(
    description=(
        "Based on the external research findings and the internal knowledge base insights "
        "for '{medical_query}', prepare an initial draft of a comprehensive research "
        "summary. "
        "The draft should include: "
        "1. A section for key external breakthroughs/developments. "
        "2. A section for how external findings relate to internal research (matches, "
        "conflicts, new angles). "
        "3. A preliminary section for potential new research avenues or adjustments to "
        "current hypotheses. "
        "4. A section for any critical warnings or uncertainties. "
        "This is a first pass, focus on getting all relevant information organized."
    ),
    expected_output=(

```

"A well-structured initial draft of a medical research summary (approx. 200-300 words) "

"for '{medical_query}', incorporating both external and internal data, ready for review."

),

agent=report_draftsman,

context=[task_research_news, task_cross_reference_kb]

)

Task 4: Supervisor Review and Finalization

task_review_and_finalize = Task(

description=(

"Review the initial draft of the research summary for '{medical_query}'. "

"Your goal is to ensure the summary is: "

"1. Comprehensive and integrates all relevant data from both external and internal sources. "

"2. Actionable, with clear recommendations for the medical research team. "

"3. Clear, concise, and professional in tone. "

"4. Free of inaccuracies or ambiguities. "

"If the draft needs refinement, re-assign the task to the 'Medical Report Draftsman' "

"with specific instructions for improvement. Otherwise, finalize the report."

),

expected_output=(

"The final, polished medical research summary for '{medical_query}', "

"ready for immediate use by the research team. This should be a concise, "

"highly actionable report (max 350 words) incorporating public and "

"proprietary data, with clear recommendations for next steps in research."

),

agent=insights_synthesizer_supervisor,

context=[task_draft_summary] # Supervisor reviews the draft

)

6. Form the Crew and Run

Form the crew

```
medical_research_crew = Crew(  
    agents=[clinical_specialist, internal_analyst, report_draftsman,  
            insights_synthesizer_supervisor],  
    tasks=[task_research_news, task_cross_reference_kb, task_draft_summary,  
            task_review_and_finalize],  
    process=Process.sequential, # Tasks run in order  
    verbose=True, # If this is giving issues, try verbose=False  
)
```

Define the medical query

medical_query_topic = "new findings on Alzheimer's disease treatments" # Example query

```
print(f"--- Starting Medical Research Crew for: {medical_query_topic} ---")  
result = medical_research_crew.kickoff(inputs={'medical_query': medical_query_topic})  
print("\n\n--- Final Research Report ---")  
print(result)
```

Presentation and Demo Script

Following is a guidance on the presentation and demo:

Title: "The AI Medical Research & Synthesis Crew: Collaborative Intelligence for Accelerated Discovery"

Objective: To showcase advanced agentic patterns in CrewAI, specifically the **Supervisor + Worker** model, to demonstrate iterative refinement and quality control in a complex medical research workflow.

(1. Introduction - The Research Challenge & Advanced Patterns)

- We're building on our CareBliss AI Navigator' concept, but now we're diving into more sophisticated **Agentic AI patterns** within CrewAI."
- "Beyond simple sequential tasks, real-world problems often require **dynamic collaboration, quality control, and iterative refinement**. This is where patterns like **Supervisor + Worker** shine."
- "Our goal remains: to rapidly synthesize information for a medical research lab, but this time, with an emphasis on **quality assurance** through intelligent oversight."

(2. The Expanded Crew - Introducing the Draftsman and Supervisor)

- "We still have our Clinical Literature Specialist and Internal Research Analyst doing their foundational work."
- "**Agent: Medical Report Draftsman (The Worker)**: This is our 'junior' analyst. Their job is to take all the raw information and produce a *first draft* of the research summary. They focus on getting all the facts in place, but not necessarily on perfection."
- "**Senior Research Insights Supervisor (The Supervisor)**: This agent does a supervisory role. Its primary goal is to **review the Draftsman's output**, identify any gaps, areas for improvement, or strategic insights that need more emphasis, and then **delegate back** to the Draftsman for revisions, or finalize the report itself if it's perfect."
- "Show the new agent definitions, highlighting `allow_delegation=True` for the Supervisor and `allow_delegation=False` for the Draftsman (as it's just doing the work)."

(3. The Iterative Workflow: Supervisor + Worker in Action)

- "Let's trace the flow of tasks, paying close attention to the new synthesis stage:"
 1. **task_research_news (Clinical Lit Specialist)**: Gathers external data.
 2. **task_cross_reference_kb (Internal Analyst)**: Gathers internal data and cross-references.
 3. **task_draft_summary (Medical Report Draftsman - Worker)**: Takes outputs from steps 1 & 2. "This is where our worker agent produces the initial summary. It's a foundational step."
 4. **task_review_and_finalize (Senior Research Insights Supervisor - Supervisor)**: "This is the critical step demonstrating the Supervisor pattern. The Supervisor agent receives the Draftsman's initial summary. It then critically evaluates it."

- "In a more advanced setup, the Supervisor could have a built-in 'critique' tool or a more complex prompt to assess quality. If it finds deficiencies, it can (through CrewAI's delegation mechanism or simply by generating instructions in its thought process for the next step) effectively 'send back' the task to the Draftsman with specific instructions for revision."
- "For this demo, the Supervisor's task description clearly states its role in reviewing and potentially delegating refinement, though CrewAI's current Process.sequential might not show a literal 'loop' within this single kickoff call. However, the *intent* of iterative refinement is captured in the Supervisor's goal and task description."
- "The Supervisor's final action is to either approve and present the refined draft or perform final polish to generate the ultimate report."
- "Show the task definitions, emphasizing the context flow from initial research to the Draftsman, and then the Draftsman's output as context for the Supervisor."

(4. Executing the Research Query & Observing Collaboration)

- "We'll run our crew for a new query: 'new findings on Alzheimer's disease treatments'."
- **Run the entire script.**
- *As the script runs, narrate the verbose output, specifically focusing on the new interactions:*
 - "Observe the Clinical Literature Specialist and Internal Research Analyst doing their work, as before."
 - "Now, watch for the Medical Report Draftsman to activate. You'll see its thought process as it tries to compile the initial summary from the previous steps."
 - "Then, the Senior Research Insights Supervisor will take over. Its thought process will involve evaluating the draft. This is where, if this were a more complex CrewAI setup or a multi-run loop, you might see it explicitly state, 'This draft needs improvement on X, Y, Z, delegating back to Draftsman,' then the Draftsman would re-engage."
 - "For this sequential demo, the Supervisor will perform its review and refinement *within its own task execution*, representing the final quality control step."

- "The verbose output should clearly show the Draftsman's *initial attempt* and then the Supervisor's *review and finalization* steps, demonstrating the separation of drafting and quality assurance."

(5. Reviewing the Final Output & The Power of Agentic Patterns)

- "Review the final result."
- "**Compare this output's quality and structure to what a single 'Synthesizer' might have produced without the dedicated drafting and supervisory layers.** Point out if it's more refined, clearer, or better organized due to the two-stage synthesis process."
- **Emphasize the value of these patterns:**
 - "This **Supervisor + Worker** pattern (and others like hierarchical teams) is crucial for complex, high-stakes tasks."
 - "It allows for **specialization** of labor (drafting vs. reviewing/refining)."
 - "It provides **built-in quality control** and a mechanism for **iterative improvement**, preventing single-agent failures or less-than-optimal outputs."
 - "It mirrors how human teams collaborate: an expert drafts, a senior expert reviews and refines."
 - "Combined with powerful LLMs from **Amazon Bedrock** and robust **CrewAI tools**, these agentic patterns enable us to build highly reliable, sophisticated, and intelligent systems for critical domains like medical research."

Conclusion

"This is a step towards truly autonomous and resilient AI systems that can tackle challenges far beyond what a single LLM call could achieve."